SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

CAP Services – E-Commerce Platform for Computer Sales & Services

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1. Problem Statement

The proposed system is a web-based e-commerce platform developed for **CAP Services**, designed for the sale, delivery, and servicing of computers and related accessories.

The platform will support a wide range of products including computers, laptops, printers, accessories (keyboard, mouse, RAM, motherboard, CPU, processors), printer consumables (A4, thermal, dot-matrix papers), cartridges, toners, and CCTV systems along with maintenance services.

In addition to product sales, the system will enable service booking and maintenance scheduling. The platform will support online payment methods such as UPI, credit/debit cards, and net banking. Cash payment will be available only for store pickup and service completion.

The system will initially be deployed on an existing GoDaddy-hosted Linux server, with future plans to migrate to a company-owned server infrastructure. The system should support domain retention or migration as required.

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2. Target Users

Customers (End Users)

Includes households, offices, institutions, and organizations.

Capabilities:

- Browse and purchase products
- Book service appointments
- Provide location details for delivery/service
- Make payments
- Submit and view reviews and feedback

Staff Users

Operational users responsible for delivery, pickup, and servicing.

Capabilities:

- View assigned tasks

- Access customer location details
- Navigate to customer location using Google Maps
- Contact customers
- Update task status
- Maintain task history

Admin Users

System administrators managing overall operations.

Capabilities:

- Monitor business activities
- Manage staff accounts and assignments
- Manage products, services, and inventory
- Handle orders and service bookings
- Analyze feedback and reviews
- Detect and act on suspicious or fraudulent activities
- Access reports and analytics

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3. System Modules

Accounts Module

- Unified login system for all user types
- Role-based authentication and redirection
- User registration and session management

Store / Shop Module (Core Backbone)

- Product catalog and categorization
- Cart and checkout system
- Payment processing integration
- Order lifecycle management
- Review and rating system
- Fraud detection and activity flagging

- Central coordination between all modules

User Module

- Product browsing and filtering
- Order placement and tracking
- Service booking system
- Review and feedback submission
- Profile and order history management

Staff Module

- Task dashboard for delivery, pickup, and servicing
- Task assignment visibility
- Status updates (pending, in progress, completed)
- Navigation support
- Task history tracking

Admin Module

- Administrative dashboard with analytics
- Staff management (add, assign, monitor)
- Product and service management
- Order and booking management
- Fraud detection and moderation
- Feedback and review monitoring

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4. Location and Navigation System

The system shall provide a flexible location input mechanism to assist staff in reaching customers efficiently.

Mandatory inputs:

- Place or area name
- Contact number

Optional inputs:

- Landmark
- Additional notes or instructions
- Google Maps location link

Navigation priority:

1. Google Maps link
2. Landmark
3. Place name

The system shall:

- Provide a one-click navigation option using Google Maps
- Allow staff to directly contact customers when required
- Support manual coordination in case of unclear locations

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5. Functional Enhancements

Inventory Management

- Inventory shall be automatically updated after every successful purchase
- The system shall prevent overselling of products
- Admin shall be able to view and update inventory manually

Notification System

Notifications shall be provided via:

- SMS
- Email

Triggers include:

- Order confirmation
- Payment confirmation
- Service booking confirmation
- Task assignment to staff
- Order and service status updates

Future enhancement:

- Mobile application with push notifications

Business Structure

- The system shall initially support a single branch
- The system shall be designed to support multi-branch expansion in the future

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6. Non-Functional Requirements

Security

- Strong input validation
- Secure authentication and session management
- Protection against fraudulent activities
- Secure payment processing

Performance

- Efficient handling of large volumes of data
- Fast response time

Usability

- Responsive and adaptive user interface
- Smooth workflow for all user types

Reliability

- Stable system operation with minimal downtime
- Proper error handling

Logging and Auditing

- All major system activities shall be recorded, including:
    - Orders
    - Payments
    - Staff actions
    - Admin actions

7. Constraints

- Must be compatible with GoDaddy Linux hosting environment
- Must be completed within a limited timeframe
- Preferred technologies include:
    - Python frameworks (Django, Flask, FastAPI)
    - PHP frameworks (Laravel, Symfony)
    - JavaScript-based stacks (Node.js)
- Avoid use of paid frameworks and licensed APIs

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8. Success Criteria

Customer Perspective

- Smooth browsing, purchasing, and service booking experience
- Secure and simple payment process
- Clear delivery and store pickup instructions

Staff Perspective

- Efficient task assignment and management
- Easy navigation to customer locations
- Reliable communication with customers
- Accurate task tracking and history

Admin Perspective

- Full control over system operations
- Efficient management of users, staff, products, and services
- Ability to monitor and respond to fraudulent activities
- Access to business insights and reports

System Perspective

- Seamless integration of all modules

- Automatic inventory updates
- Reliable notification system
- Scalability for future multi-branch expansion